Used Car Price Prediction

Data preparation and regression baselines for used-car listings

PROJECT SUMMARY

This project examines the supplied used-car dataset and builds two regression baselines using vehicle attributes such as brand, year, engine size, fuel type, transmission, mileage, condition, and model.

DATA AND SCOPE

2,500 records and 10 source columns. The workflow normalizes category text, removes duplicates, converts numeric fields, creates car age relative to the current year, and excludes invalid prices and implausible ages.

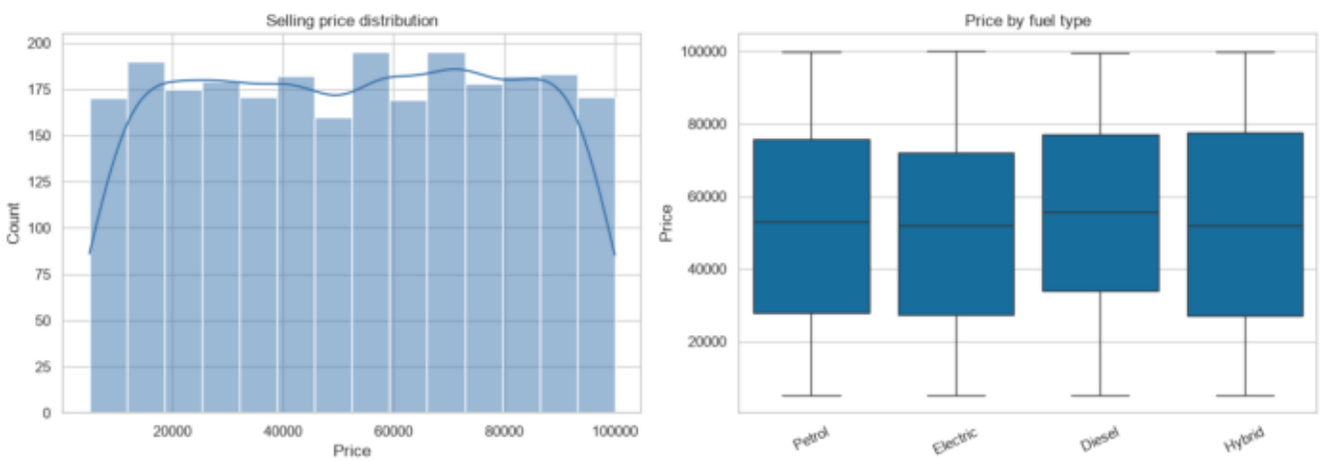
ANALYSIS APPROACH

Explored price distribution, price by fuel type, price versus age, and numeric correlations. Used a train/test split, median imputation and scaling for numeric fields, most-frequent imputation and one-hot encoding for categorical fields. Compared Ridge Regression and Random Forest with MAE, RMSE, and R-squared.

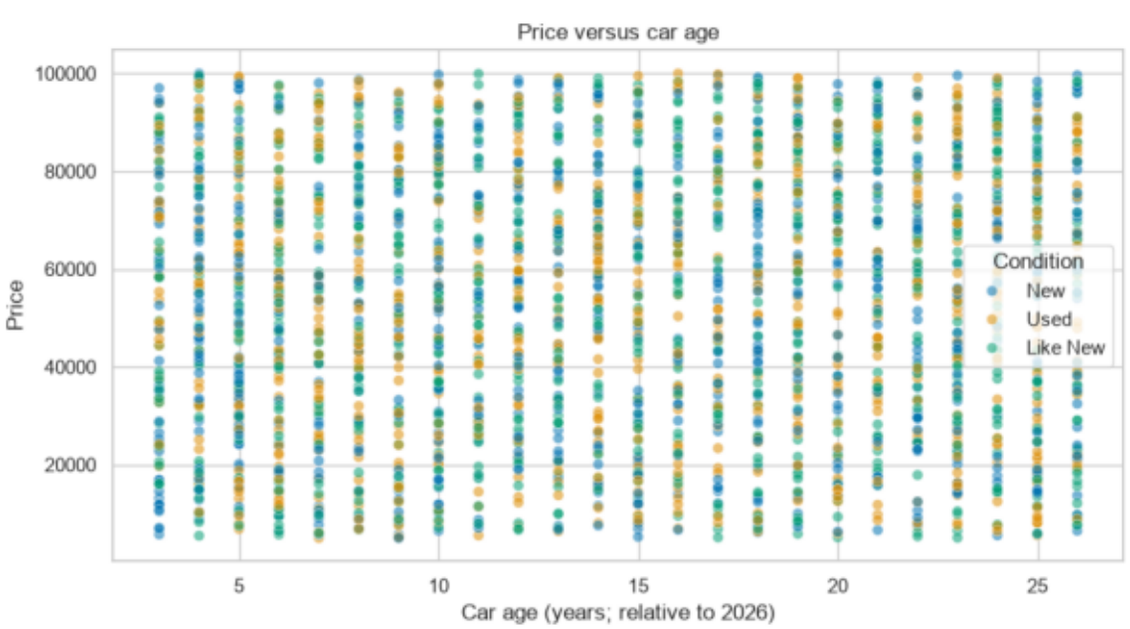
Prepared from the executed project notebook and supplied data.

RESULTS AND EVIDENCE

Ridge Regression was the better of the two tested models on the held-out split, with MAE 23,873.875, RMSE 27,788.760, and R-squared -0.019. The random forest had MAE 24,296.377, RMSE 28,208.886, and R-squared -0.050. Negative R-squared means neither baseline beat predicting the test-set mean on this split; this is an honest baseline, not a reliable pricing solution.



Selected notebook visualization



Selected notebook visualization

INTERPRETATION AND REPRODUCTION

Limitations and responsible use

The supplied synthetic-looking data show weak learnable signal for these features and this split. Better sourced listings, currency/market context, and validation of price quality are needed before using the model for pricing decisions.

Reproduce the analysis

Open used_car_price_prediction.ipynb in Jupyter and run all cells. Keep car_price_prediction_.csv in this folder.

The notebook includes the complete EDA, charts, modeling code, metric tables, and cell outputs.

Interpret model scores in the context of the dataset and holdout design. Observational patterns and predictive associations do not establish causal effects.